

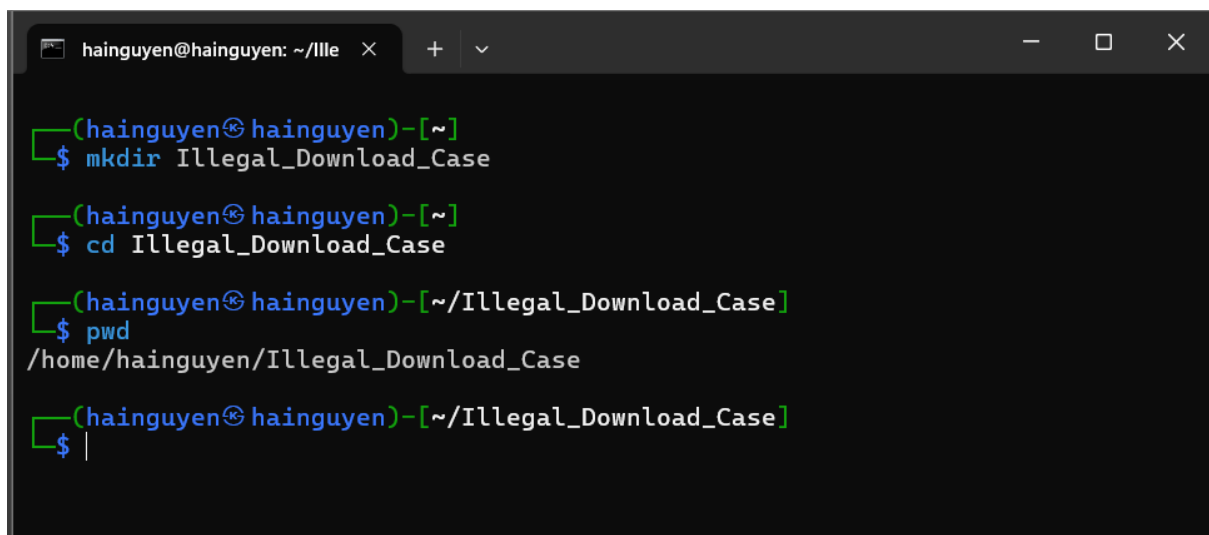
Lab 4: Disk Image and Partitions

What You Need for this lab

- Install Virtualbox : <https://www.virtualbox.org/wiki/Downloads>
- Install Kali 2021.4. : <https://old.kali.org/kali-images/kali-2021.4/>
 - Notes: Suggest You configure the disk size of Kali VM 80G because the size of each leakage cases image is 30G+
- Disk image : https://www.dropbox.com/s/1fop1ooadb2yshu/Disk_Image_ID-20210327.001

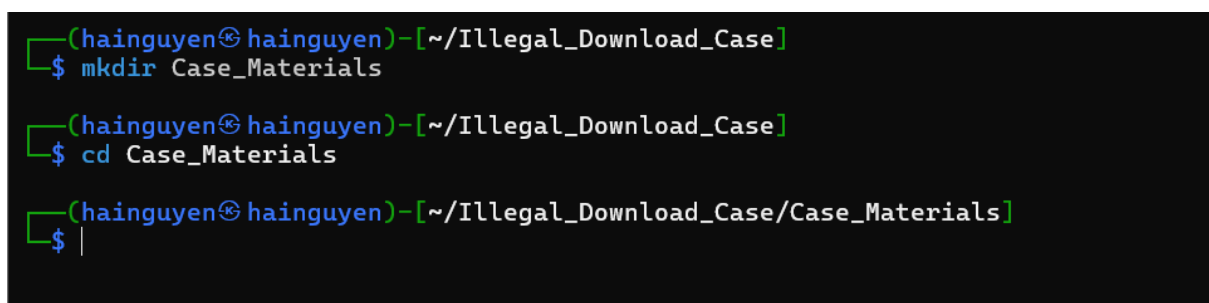
1. Verify the integrity of the disk image

- Create Lab Folder



```
hainguyen@hainguyen: ~/Ille x + v
(hainguyen@hainguyen)-[~]
$ mkdir Illegal_Download_Case
(hainguyen@hainguyen)-[~]
$ cd Illegal_Download_Case
(hainguyen@hainguyen)-[~/Illegal_Download_Case]
$ pwd
/home/hainguyen/Illegal_Download_Case
(hainguyen@hainguyen)-[~/Illegal_Download_Case]
$ |
```

- Download Case Materials



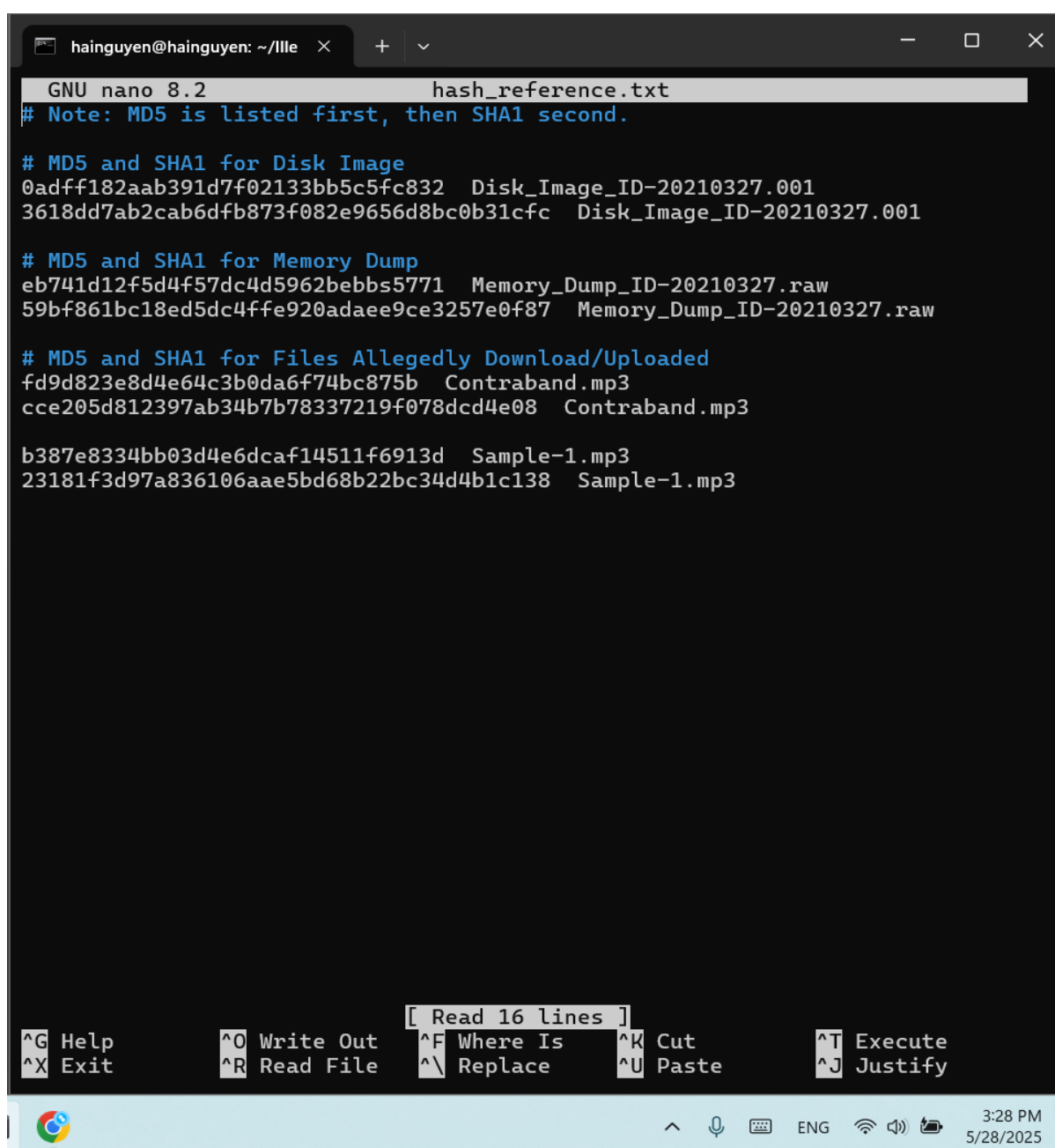
```
(hainguyen@hainguyen)-[~/Illegal_Download_Case]
$ mkdir Case_Materials
(hainguyen@hainguyen)-[~/Illegal_Download_Case]
$ cd Case_Materials
(hainguyen@hainguyen)-[~/Illegal_Download_Case/Case_Materials]
$ |
```

- Use `wget` to download disk image. (about 30GB)

```
(hainguyen@hainguyen)-[~/Illegal_Download_Case/Case_Materials]
$ wget -q https://www.dropbox.com/s/1fop1oadb2yshu/Disk_Image_ID-20210327.001
(hainguyen@hainguyen)-[~/Illegal_Download_Case/Case_Materials]
$
```

- Record Hash Information
- Open a text file using the text editor Nano:

```
(hainguyen@hainguyen)-[~/Illegal_Download_Case/Case_Materials]
$ nano hash_reference.txt
```



```
GNU nano 8.2 hash_reference.txt
# Note: MD5 is listed first, then SHA1 second.

# MD5 and SHA1 for Disk Image
0adff182aab391d7f02133bb5c5fc832 Disk_Image_ID-20210327.001
3618dd7ab2cab6dfb873f082e9656d8bc0b31cfc Disk_Image_ID-20210327.001

# MD5 and SHA1 for Memory Dump
eb741d12f5d4f57dc4d5962bebb55771 Memory_Dump_ID-20210327.raw
59bf861bc18ed5dc4ffe920adaee9ce3257e0f87 Memory_Dump_ID-20210327.raw

# MD5 and SHA1 for Files Allegedly Download/Uploaded
fd9d823e8d4e64c3b0da6f74bc875b Contraband.mp3
cce205d812397ab34b7b78337219f078dcd4e08 Contraband.mp3

b387e8334bb03d4e6dc4f14511f6913d Sample-1.mp3
23181f3d97a836106aae5bd68b22bc34d4b1c138 Sample-1.mp3

[ Read 16 lines ]
^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify
```

- Install Necessary Software

- ☐ Hashdeep
- ☐ Md5deep

sudo apt install hashdeep

Step 2.

Make sure you are in the lab folder

```
(hainguyen@hainguyen)-[~]  
$ cd Illegal_Download_Case  
  
(hainguyen@hainguyen)-[~/Illegal_Download_Case]  
$ |
```

Generate an MD5 and SHA1 hash of the disk image. These tools will compare the MD5 and/or SHA1 hash of the disk image to the MD5 and/or SHA1 hash in the 'hash_reference.txt' file.

Commands

- `md5deep <disk image> -bewM <file that contains file names and hash codes>`
- `sha1deep <disk image> -bewM <file that contains file names and hash codes>`
- Note: You would replace <disk image> with the file path to the disk image. The same applies to anything else contained in between '<>'.
 - Use MD5deep to verify the MD5 hash of the disk image.

```
(hainguyen@hainguyen)-[~/Illegal_Download_Case]  
$ md5deep Case_Materials/Disk_Image_ID-20210327.001 -bewM Case_Materials/h  
ash_reference.txt  
md5deep: Case_Materials/hash_reference.txt: No hash found in line 1  
md5deep: Case_Materials/hash_reference.txt: No hash found in line 2  
md5deep: Case_Materials/hash_reference.txt: No hash found in line 3  
md5deep: Case_Materials/hash_reference.txt: No hash found in line 5  
md5deep: Case_Materials/hash_reference.txt: No hash found in line 6  
md5deep: Case_Materials/hash_reference.txt: No hash found in line 7  
md5deep: Case_Materials/hash_reference.txt: No hash found in line 8  
md5deep: Case_Materials/hash_reference.txt: No hash found in line 9  
82369df18db442a8d69f5ed81858d634 Disk_Image_ID-20210327.001 matched Disk_Im  
age_ID-20210327.001  
  
(hainguyen@hainguyen)-[~/Illegal_Download_Case]  
$ |
```

- Use SHA1deep to verify the MD5 hash of the disk image.

```
(hainguyen@hainguyen)~[/Illegal_Download_Case]
$ sha1deep Case_Materials/Disk_Image_ID-20210327.001 -bewM Case_Materials/
hash_reference.txt
sha1deep: Case_Materials/hash_reference.txt: No hash found in line 1
sha1deep: Case_Materials/hash_reference.txt: No hash found in line 2
sha1deep: Case_Materials/hash_reference.txt: No hash found in line 3
sha1deep: Case_Materials/hash_reference.txt: No hash found in line 4
sha1deep: Case_Materials/hash_reference.txt: No hash found in line 6
sha1deep: Case_Materials/hash_reference.txt: No hash found in line 7
sha1deep: Case_Materials/hash_reference.txt: No hash found in line 8
sha1deep: Case_Materials/hash_reference.txt: No hash found in line 9
cfla5bef6a743227bf79d5e55ffe90359fc08135 Disk_Image_ID-20210327.001 matched
Disk_Image_ID-20210327.001

(hainguyen@hainguyen)~[/Illegal_Download_Case]
$
```

2. Identify the OS of the system as well as its name, accounts, and partitions.

- How to get help for *fdisk*.

```
(hainguyen@hainguyen)~[/Illegal_Download_Case]
$ fdisk -h

Usage:
fdisk [options] <disk>          change partition table
fdisk [options] -l [<disk>...] list partition table(s)

Display or manipulate a disk partition table.

Options:
-b, --sector-size <size>      physical and logical sector size
-B, --protect-boot            don't erase bootbits when creating a new label
-c, --compatibility[=<mode>]  mode is 'dos' or 'nondos' (default)
-L, --color[=<when>]          colorize output (auto, always or never)
                              colors are enabled by default
-l, --list                    display partitions and exit
-x, --list-details            like --list but with more details
-n, --noauto-pt               don't create default partition table on empty
                              devices
-o, --output <list>           output columns
-t, --type <type>             recognize specified partition table type only
-u, --units[=<unit>]          display units: 'cylinders' or 'sectors' (default)
-s, --getsz                   display device size in 512-byte sectors [DEPRECATED]
--bytes                       print SIZE in bytes rather than in human readable format
--lock[=<mode>]               use exclusive device lock (yes, no or nonblock)
-w, --wipe <mode>            wipe signatures (auto, always or never)
-W, --wipe-partitions <mode> wipe signatures from new partitions (auto, always or never)
```

- Use *fdisk* to get the disk image's partition table.

```

(hainguyen@hainguyen)~[/Illegal_Download_Case]
$ cd Case_Materials

(hainguyen@hainguyen)~[/Illegal_Download_Case/Case_Materials]
$ |

(hainguyen@hainguyen)~[/Illegal_Download_Case/Case_Materials]
$ fdisk -l Disk_Image_ID-20210327.001
Disk Disk_Image_ID-20210327.001: 496.11 MiB, 520208384 bytes, 1016032 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x8afa8be3

Device            Boot    Start        End    Sectors    Size Id Type
Disk_Image_ID-20210327.001p1 *      2048        104447    102400     50M 7 HPFS/N
Disk_Image_ID-20210327.001p2      104448    61890501    61786054    29.5G 7 HPFS/N
Disk_Image_ID-20210327.001p3    61890560    62910463    1019904     498M 27 Hidden

(hainguyen@hainguyen)~[/Illegal_Download_Case/Case_Materials]
$ |

```

Volume offset #s (in sectors):

- Partition 1 – 2048
- Partition 2 – 104448
- Partition 3 – 61890560
- How to get help for *fsstat*

```

(hainguyen@hainguyen)~[/Illegal_Download_Case/Case_Materials]
$ fsstat
Missing image name
usage: fsstat [-tv] [-f fstype] [-i imgtype] [-b dev_sector_size] [-o imgoffset] image
  -t: display type only
  -i imgtype: The format of the image file (use '-i list' for supported types)
  -b dev_sector_size: The size (in bytes) of the device sectors
  -f fstype: File system type (use '-f list' for supported types)
  -o imgoffset: The offset of the file system in the image (in sectors)
  -P pooltype: Pool container type (use '-P list' for supported types)
  -B pool_volume_block: Starting block (for pool volumes only)
  -v: verbose output to stderr
  -V: Print version
  -k password: Decryption password for encrypted volumes

(hainguyen@hainguyen)~[/Illegal_Download_Case/Case_Materials]
$ |

```

- Use *fsstat* to get file system details.

```

(hainguyen@hainguyen)~[/Illegal_Download_Case/Case_Materials]
$ fsstat -o 2048 Disk_Image_ID-20210327.001
FILE SYSTEM INFORMATION
-----
File System Type: NTFS
Volume Serial Number: 18EC42BBEC4292C4
OEM Name: NTFS
Volume Name: System Reserved
Version: Windows XP

METADATA INFORMATION
-----
First Cluster of MFT: 4266
First Cluster of MFT Mirror: 2
Size of MFT Entries: 1024 bytes
Size of Index Records: 4096 bytes
Range: 0 - 256
Root Directory: 5

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 4096
Total Cluster Range: 0 - 12798
Total Sector Range: 0 - 102398

```

```

(hainguyen@hainguyen)~/Illegal_Download_Case/Case_Materials]
$ fsstat -o 104448 Disk_Image_ID-20210327.001
FILE SYSTEM INFORMATION
-----
File System Type: NTFS
Volume Serial Number: E8DE4350DE4315EA
OEM Name: NTFS
Version: Windows XP

METADATA INFORMATION
-----
First Cluster of MFT: 786432
First Cluster of MFT Mirror: 2
Size of MFT Entries: 1024 bytes
Size of Index Records: 4096 bytes
Range: 0 - 226304
Root Directory: 5

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 4096
Total Cluster Range: 0 - 7723255
Total Sector Range: 0 - 61786052

```

```

(hainguyen@hainguyen)~/Illegal_Download_Case/Case_Materials]
$ fsstat -o 61890560 Disk_Image_ID-20210327.001
FILE SYSTEM INFORMATION
-----
File System Type: NTFS
Volume Serial Number: 9E46F86046F83A9B
OEM Name: NTFS
Version: Windows XP

METADATA INFORMATION
-----
First Cluster of MFT: 42496
First Cluster of MFT Mirror: 2
Size of MFT Entries: 1024 bytes
Size of Index Records: 4096 bytes
Range: 0 - 256
Root Directory: 5

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 4096
Total Cluster Range: 0 - 127486
Total Sector Range: 0 - 1019902

```

Partition 1

File System: **NTFS**

Serial Number: **18EC42BBEC4292C4**

Partition 2

File System: **NTFS**

Serial Number: **E8DE4350DE4315EA**

Partition 3

File System: **NTFS**

Serial Number: **9E46F86046F83A9B**

- Using *fdisk* and *fsstat*, we obtained this information:

Partition Table			MS-DOS				
Partition	Flag	Start	End	Sectors	Size	File System	Serial #
1 st Partition – System Reserved	Boot	2048	104447	102400	50 MB	NTFS	18EC42BBEC4292C4
2 nd Partition	-	104448	61890501	61786054	29.5 GB	NTFS	E8DE4350DE4315EA
3 rd Partition	-	61890560	62910463	1019904	498 MB	NTFS/Hidden NTFS WinRe	9E46F86046F83A9B

Please explain the parameters in the table ?

Parameter	Explanation
Partition	Indicates the specific partition on the storage device. - 1st Partition – System Reserved: Often created by Windows during installation and used for boot files. - 2nd Partition: Typically the main OS or user data partition. - 3rd Partition: May be recovery or system-specific partition.
Flag	Describes the role of the partition: - Boot: This partition is marked as bootable; the system looks here to start the OS. - “-” (dash): No special flag.
Start	The starting sector number on the disk where the partition begins. Disk sectors are the smallest addressable unit of storage (typically 512 bytes or 4 KB).
End	The ending sector number on the disk where the partition ends.
Sectors	The total number of sectors the partition spans. Calculated as: End - Start + 1.
Size	The total size of the partition, usually calculated by multiplying the number of sectors by the sector size (often 512 bytes). Converted to MB or GB for readability.
File System	The formatting system used to organize files. Examples: - NTFS: New Technology File System (used by Windows). - NTFS/Hidden NTFS/WinRe: This suggests a hidden NTFS partition often used for Windows Recovery Environment.
Serial #	A unique identifier for the partition, generated when the partition is created. Used internally by the system.

YOU MUST SUBMIT A FULL-SCREEN IMAGE FOR FULL CREDIT!

Save the document with the filename "**YOUR NAME Lab 4.pdf**", replacing "YOUR NAME" with your real name.

Email the image to the instructor as an attachment to an e-mail message. Send it to: **xxx@fe.edu.vn** with a subject line of "**Lab 4 From YOUR NAME**", replacing "YOUR NAME" with your real name.